

An evacuation path planning method for multi-hazard accidents in chemical industries based on risk perception

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ARTICLE INFO

Keywords:

Emergency evacuation
Multi-hazard accident
Risk analysis
Chemical industry
D* algorithm

ABSTRACT

Evacuation path planning is an important research topic in emergency management of chemical accidents. Facing the challenges of the continuous increase in the frequency of multi-hazard chemical accidents including fires, explosions, and toxic releases, the requirement to improve the evacuation path planning method considering the dynamic and multi-hazard characteristics of the accidents has become increasingly urgent. In order to address the shortcomings of the static, single-hazard, and passive evacuation path planning methods, this paper proposes an evacuation path planning method for multi-hazard accidents in chemical industries based on advance and on-site risk perception. This method applies dynamic and comprehensive risk analysis as the equivalent distance criterion to improve the D* path planning algorithm. A chemical park in Guangzhou, China, is adopted as a case study to verify the effectiveness of the proposed method and quantitatively demonstrate the effect of the method in improving evacuation efficiency. This paper further discusses the flexibility of the proposed method in considering environmental factors, evacuation preferences, and emergency response requirements. The evacuation path planning method proposed in this paper is expected to contribute to the optimization of emergency management of multi-hazard accidents and provide guidance for the enhancement of safety and loss prevention in chemical industries.

1. Introduction

With the development of the chemical industry, the frequency of multi-hazard accidents involving fires, explosions, and toxic releases has been continually increasing. The uncertain and unpredictable characteristics of multi-hazard accidents pose threats to both individual and societal safety within the industrial area and its vicinity [1]. A typical multi-hazard chemical accident happened in Xiangshui, China in 2019, where a fire triggered explosions and toxic releases, causing 78 fatalities and 716 injuries [2]. The optimization of emergency management methods and strategies for multi-hazard accidents is a cutting-edge topic in chemical safety research.

Evacuation planning is an important research topic in emergency management for chemical accidents. An improper decision during the evacuation process can lead to severe accident consequences. For example, in an accident that occurred in Yingkou, China in 2023, an unexpected explosion led to nine casualties during the evacuation process [3]. In order to mitigate the risk of casualty during the evacuation

process, scholars recently proposed a variety of evacuation planning methods for chemical accidents.

In the previous research, automatic pathfinding algorithms such as A-star and Dijkstra algorithms were the most commonly used methods for accident evacuation path planning [4–8]. Other methods such as graph theory [8], cellular automaton [9], artificial intelligence [10], and extreme learning machine [11] have also been adopted by scholars. Scholars have also proposed different criteria for evacuation path planning, such as FMEA (Failure mode and effects analysis) [12], computational fluid dynamics simulation [4,10,13], decision-making matrix [14], “ALARP” guidelines [15], dynamic risk assessment [8,16], and multi-objective optimization technique [17]. Chemical fires are the most concerning accident scenarios of evacuation planning [7,18,19]. Several research considered the multi-hazard scenarios involving fires and toxic releases [9,20]. Other literature presented practical software tools for evacuation path planning, such as Pathfinder, LEGION, and MATSim [14,21,22]. In recent research, Liu et al. [23] applied similar intelligent methods to conduct research on emergency

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rescue planning in metro systems. Chen et al. [24] studied the evacuation process considering the indoor fire accident risk.

Facing the challenges of dynamic and multi-hazard characteristics of chemical accidents, the shortcomings of the previous research on accident evacuation planning need to be addressed. First is the lack of consideration of the dynamic development of chemical accidents. Although the dynamic path planning algorithms (e.g., dynamic A-star algorithm) can consider the dynamic changes of the consequences of independent accidents [18], there is still a lack of consideration of the dynamic triggering process (domino effect) during the development of multi-hazard accidents [9]. Second is the lack of analysis of explosion accidents. Explosions can lead to out-of-control and unpredictable accident development, but these were rarely considered in the previous research on accident evacuation path planning.

Another problem is the lack of risk pre-perception in accident evacuation path planning, or in other words, the advance recognition and assessment of accident risks [8]. Most of the previous path planning algorithms determine the optimal evacuation strategies based on the real-time development of accidents and the on-site risk identification and assessment, or the risk field-perception. However, the lack of prediction of the dynamic changes of the accident risk can lead evacuees to be trapped in potentially high-risk areas, and suffer higher risk during the evacuation process. The above shortcomings limit the applicability of the previous methods of evacuation planning for multi-hazard accidents. Focusing on the dynamic and multi-hazard characteristics of the chemical accident, the optimization of evacuation path planning methods is expected to improve evacuation efficiency.

This paper aims to propose an evacuation path planning method for multi-hazard accidents in chemical industries. The method is based on the improved D* (dynamic A-star) algorithm and can consider the dynamic, multi-hazard, and domino effect characteristics of fire, explosion, and toxic release accidents. This method combines the time consumed and accident risk of the evacuation process as a cumulative individual risk (CIR) indicator, served as the equivalent distance criterion for the D* path planning algorithm. Moreover, this method uses Monte Carlo simulation to conduct data synthesis of the multi-hazard accidents. The comprehensive accident risk is further analyzed as the risk pre-perception basis of the evacuation path planning method.

The major innovation of this method is the capability of dynamic planning and adjustment of the optimal evacuation path according to the development of multi-hazard domino accidents, and the improvement of the cost functions with the consideration of risk analysis and perception. Risk perception refers to people's beliefs, attitudes, judgments, and feelings toward risk, which frequently appears in societal research [25]. This paper adopts this concept to describe the ability of the intelligent automatic path planning method to perceive accident risks like humans. The algorithm can subjectively "feel" the on-site accident risks and predict potential subsequent accident risks like real evacuees, and make real-time judgments and decisions based on risk perception. Under the guidance of the method, evacuees can determine the escape path according to their field perception of the real-time risk and the pre-perception of the comprehensive risk, and can also adjust the path planning algorithm according to their preferences. This method also has the potential to optimize emergency response strategies such as the approach timing and path according to the risk tolerance of emergency responders (firefighters, medical staff, etc.). The application and promotion of the proposed method are expected to contribute to the enhancement of safety and loss prevention in chemical industries.

In this paper, Section 2 introduces the basis, logic, and steps of the proposed method, especially emphasizing the improved D* algorithm based on dynamic risk analysis, and the comprehensive risk perception based on data synthesis. Section 3 adopts a typical multi-hazard accident as a case study to verify the effectiveness of the proposed method, and discuss the efficiency improvement compared to other evacuation path planning methods. Section 4 discusses the flexibility of the method. This section introduces the influence of environmental factors and evacuee's

preferences on the path planning method, and the applicability of the method to the optimization of emergency response strategies. Section 5 provides some concluding remarks.

2. Methodology

The evacuation path planning method proposed in this paper is based on two major improvements to the D* algorithm. First is the application of the equivalent distance criterion based on the dynamic risk analysis of multi-hazard accidents. Second, is the application of the comprehensive risk perception to improve the algorithm of evacuation cost. The method can be divided into five steps according to different stages: scene construction, dynamic risk analysis, comprehensive risk analysis, optimal path planning, and CIR calculation. This section aims to introduce the two concepts, steps, and innovations of the method in detail.

2.1. Concept 1: equivalent distance criterion

D* is a heuristic path planning algorithm, the core of which is to dynamically adjust the planning path according to real-time updates to the total cost (sum of actual cost and estimated cost) [22]. Most applications of the algorithm targeted the shortest path, with both the actual and estimated costs using Euclidean distance or Manhattan distance as criteria [26,27]. Nevertheless, evacuation after the occurrence of multi-hazard accidents should not only aim at the shortest evacuation path and time, but also consider the accident risk during the evacuation process. To achieve this optimization goal, scholars have proposed indicators such as road risk and accident risk, and applied risk matrix to conduct a comprehensive evaluation [14,28]. However, the unnormalized evaluation indicators make the optimization of path planning still a difficult problem to solve.

The proposed evacuation path planning method uses the concept of cumulative individual risk (CIR) as the equivalent distance to replace the criteria of actual and estimated cost. Individual risk is a widely used concept in risk management of chemical accidents, which can comprehensively consider the accident consequence ρ and frequency φ . Cumulative risk is a commonly used evaluation indicator in research fields such as occupational health [29] and medical treatment [30]. The CIR indicator is based on the concept of individual risk in multi-hazard accidents, referring to the accumulation of individual risk suffered by evacuees during the evacuation period. This improvement allows the path planning method to comprehensively consider factors such as evacuation time consumed and multi-hazard accident risk.

$$CIR(T) = \int_0^T \left(\sum_i^n \rho_{i,t} \varphi_{i,t} \right) dt \quad (1)$$

This equation shows the CIR as a function of the total evacuation time T . n refers to the total number of hazards. $\rho_{i,t}$ and $\varphi_{i,t}$ represent the consequence (human death rate) and frequency of hazard i in real time t , respectively.

The calculation of CIR is based on the dynamic risk analysis of multi-hazard accidents. By applying consequence analysis models [31] and damage analysis models [32], the dynamic distribution of individual risks can be quantitatively calculated based on the development of the domino process in fire, explosion, and toxic release accidents. Several references have introduced the steps of dynamic risk analysis of multi-hazard accidents in detail [33,34]. According to the individual risk at the evacuee's location at each moment during the evacuation process, the CIR corresponding to the evacuation path can be calculated by discrete summation.

In the evacuation path planning algorithm, the real-time individual risk of the position that the evacuee can reach in the next step is taken as the actual cost. The CIR from this position to the evacuation destination is taken as the estimated cost by modifying the heuristic function. The minimum total cost is used as the planning criterion to determine the

optimal evacuation path. Algorithm 1 shows the pseudo code of the evacuation path planning method applying equivalent distance as the criterion.

Algorithm 1 Evacuation path planning algorithm

```

1      procedure EvacuationAlgorithm
2      Load risk distribution
3      Set starting position coordinates
4      Add starting position to path vector
5      Set stop condition  $\leftarrow$  false, counter  $\leftarrow$  0
6      while  $\sim$  stopCondition
7          counter = counter + 1
8          Get the previous position from path vector
9          for each surrounding position
10             Determine actual cost  $\leftarrow$  individual risk
11             Calculate CIR from the position to each destination
12             estimated cost  $\leftarrow$  minimum CIR
13             Determine the next position with minimum total cost
14             Add next position to path vector
15             if next point reaches destination
16                 stopCondition  $\leftarrow$  true
17             end if
18         end for
19     end while
20 end procedure

```

2.2. Concept 2: comprehensive risk perception

In addition to applying dynamic risk analysis of accidents as the criterion for evacuation path planning, the proposed method also applies comprehensive risk analysis to predict possible emergencies during the evacuation process. Based on pre-perception of the comprehensive accident risk, the evacuation path planning method can consciously avoid potential high-risk areas. The field perception and the pre-perception of the accident risk jointly determine the optimal evacuation path, thereby preventing evacuees from blindly evacuating and suffering damage from unexpected emergencies.

The multi-hazard accidents in chemical industries are typical high-consequence low-frequency events. An insufficient number of actual cases leads to a lack of sample basis for data analysis. Based on the dynamic simulation of the multi-hazard domino accident process, this method applies a data synthesis method using Monte Carlo simulation (MCS) [34]. MCS is the iteration of single simulations of the dynamic development of domino accidents. A single simulation starts from the primary accident, and applies escalation probability models [35] to simulate the random process of accident development until the accident fully develops. According to the data analysis of all possible accident scenarios, MCS can calculate the comprehensive accident risk distribution of the area. Fig. 1 shows the logic of the MCS process.

The comprehensive risk analysis results obtained by MCS are the risk pre-perception basis of the evacuation path planning method. **The improvement of the D* algorithm based on risk pre-perception lies in the modification of the heuristic function.** The field perception of real-time risk and the pre-perception of comprehensive risk constitute the heuristic function of the evacuation path planning algorithm. The weights of the two (α, β) are determined by the preferences of the evacuation planner. The improvements change the target of the evacuation path planning into the minimization of the weighted average of actual (real-time) risk and potential (comprehensive) risk. The equation is shown below. More preference terms can also be added to the heuristic function according to the requirements of evacuation planners and the actual situation of the accident scene.

$$\text{estimated cost} = \alpha \times CIR(\text{field} - \text{perception}) + \beta \times CIR(\text{pre} - \text{perception}) \quad (2)$$

2.3. Steps of evacuation path planning method

Fig. 2 shows the flow of the evacuation path planning method based on risk field-perception and pre-perception. This section aims to introduce each step of the method in detail.

(1) Scene construction

This step aims to collect information on geography, environment, hazardous materials, installations, and society necessary for dynamic simulation and risk analysis of multi-hazard accidents. This step requires identifying all potential accident types in the industrial area, and preparing corresponding accident analysis models (consequence, escalation, damage analysis models, etc.) and basic data (e.g., frequency of occurrence). The reference books published by CCPS [31] and TNO [32] can be the reference for this step.

(2) Dynamic risk analysis

Section 2.1 has introduced the steps of the dynamic risk analysis. By applying the accident analysis models, this step aims to calculate the multi-hazard physical effect fields (thermal radiation, overpressure, concentration) according to the real-time development of the accident. The dynamic distribution of individual risk can be calculated based on the consequence analysis results and the data of occurrence frequency. This step is the basis of risk field-perception for the evacuation path planning method.

(3) Comprehensive risk analysis

Comprehensive risk analysis is based on the MCS-based data synthesis introduced in Section 2.2. Through the iterative simulation and data analysis of random domino processes of the multi-hazard accident, MCS can calculate the comprehensive individual risk distribution in and around the industrial area. This step is the basis of risk pre-perception for the evacuation path planning method, which can be performed

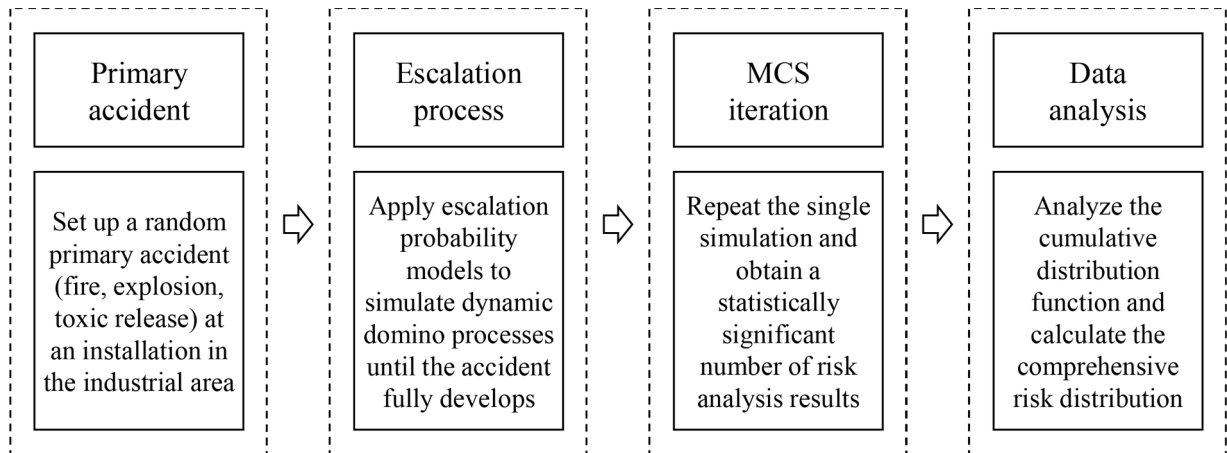


Fig. 1. The logic of the MCS process.

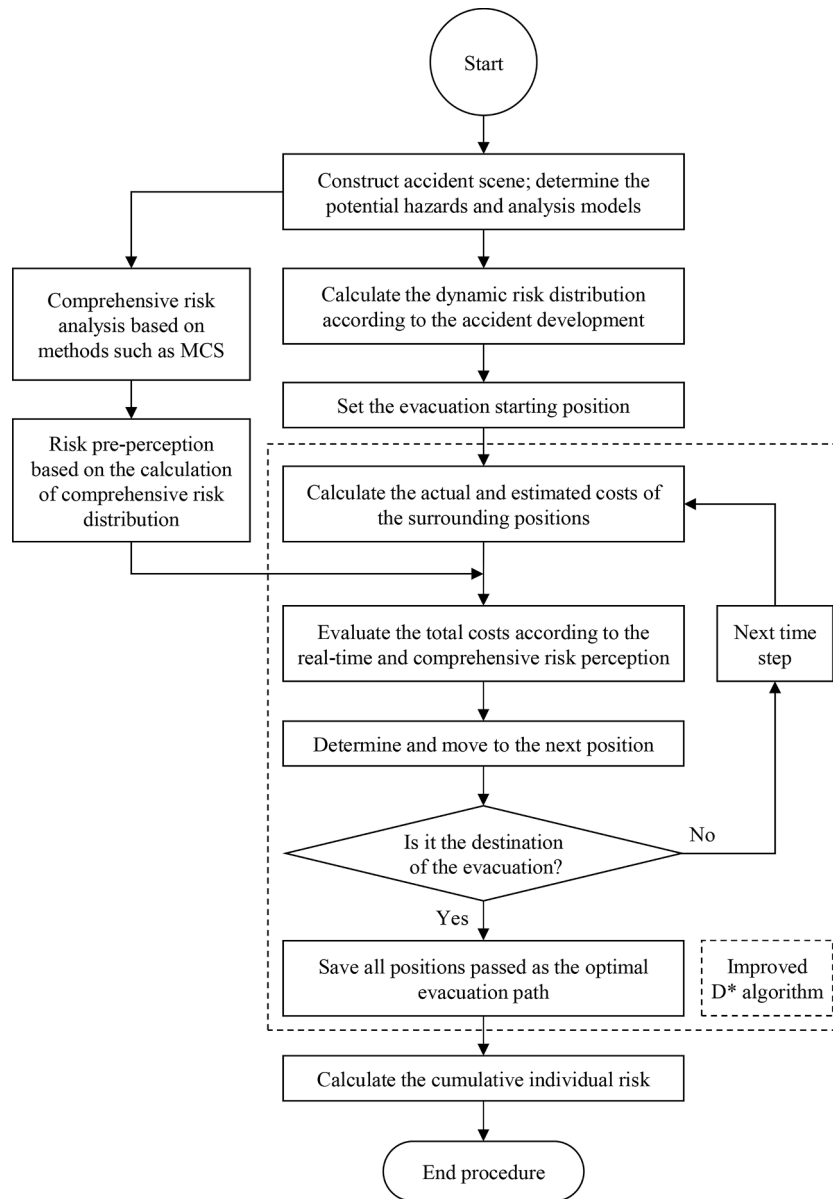


Fig. 2. The flow of the evacuation path planning method based on risk field-perception and pre-perception.

before the accident occurs.

(4) Optimal path planning

Fig. 2 shows the iterative algorithm of the evacuation path planning method. The total cost of each choice in the next step should be evaluated and judged at each moment during the evacuation until evacuees reach the destination. If evacuees have alternative destinations, the judgment should include the total cost of each choice to each destination. As shown in Eq. (2), the weights of estimated cost can be adjusted according to the risk preference of the evacuation planner. Information about the optimal evacuation path can be delivered to evacuees through real-time communication or on-site guidance.

(5) CIR calculation

This step aims to calculate the CIR suffered by evacuees based on the evacuation path planning result. CIR data that changes over time can guide evacuation planners to conduct risk evaluation, determine evacuation decisions, and formulate rescue strategies. Based on the CIR comparison of different evacuation paths, planners can optimize emergency response strategies and guide evacuees to make effective and rational evacuation decisions.

2.4. Method's advantages and shortcomings

Compared to the previous evacuation path planning methods for chemical accidents, the proposed method has advantages in aspects of multi-hazard and dynamic characteristics, risk-based criterion, practicality, and flexibility. The proposed method can consider the domino process of multi-hazard accidents in evacuation path planning, and adjust the optimal path dynamically according to the accident development. This is one of the advantages of this method compared to the evacuation path planning methods for static and independent accident scenarios.

The second advantage is that the algorithm of this method can simulate evacuees' on-site perception and prediction of accident risks and apply them as the basis for the equivalent distance criterion. The purpose of evacuation planning is to allow evacuees to leave the accident area as safely as possible, which makes it necessary to comprehensively consider the time consumed and the accident risk. The method applies the CIR indicator to improve the heuristic function, which can not only normalize the time and risk criteria, but also predict potential high-risk areas. This advantage prevents evacuees from making wrong

decisions based on subconsciousness, thereby improving evacuation efficiency.

Due to the application of accident physical effect field analysis and individual risk distribution analysis as the basis of the D* path planning algorithm, the proposed method can show more details of the evacuation process. Most previous path planning methods with the application of Dijkstra's algorithm took significant locations in the area as nodes and alternative evacuation paths as connections between nodes [6,8,18,20]. In comparison, the evacuation path planned by the proposed method can describe the real-time decisions made by evacuees in the dynamic accident development in more detail.

Another advantage of the proposed method is the flexibility of the heuristic function. In addition to the considerations of time consumed and CIR, different accident scenarios can also have specific requirements on evacuation preference (direction preference, time preference, etc.). By adding new preference terms to the heuristic function, evacuation preferences can be considered in path planning. Evacuation planners can also adjust the weights of the heuristic function according to their evaluation of the actual accident scene.

The shortcomings of the method are also nonnegligible. The proposed method only considers microscopic individual evacuation and cannot consider the crowding and congestion problems in macroscopic crowd evacuation. This method also adopts the assumption of uniform speed to simplify the evacuation process, and does not consider the impact of the accident on evacuees, such as the speed reduction caused by the accident damage. Further improvements to the evacuation path planning method will aim at addressing the above shortcomings, thereby enhancing the practical value of the proposed method.

3. CASE study

A hazardous material storage park on the coast of Guangzhou, China, is taken as a case study to verify the effectiveness of the proposed evacuation path planning method for chemical accidents.

3.1. Scene construction

Storage tanks contain gasoline, ethylene glycol, and hydrogen chloride, which can cause fire, explosion, and toxic release multi-hazard accidents. Fig. 3 shows the overlook and the plan of this chemical park. In order to reduce the complexity of the accident scene, we adopt appropriate assumptions and simplifications. Small-scale storage tanks and complicated installations such as pipelines and pumps are not

considered in this study. The types of fire, explosion, and toxic release accidents are simplified and limited to pool fire, vapor cloud explosion, and continuous release.

3.2. Dynamic risk analysis

The case study assumes a domino accident scenario in this chemical park. Fires, explosions, and toxic releases happen successively at different hypothetical time points. The primary accident is a fire accident that happens in tank #1, zone 3. The fire causes an explosion in tank #2 in zone 3 ten seconds later. The explosion causes all nearby storage tanks to be damaged and start burning. At 38 s, tanks #1 and #3 in zone 2 start releasing toxic gas. High-temperature environment causes tank #1 in zone 5 to explode at 92 s, which also ignites the nearby storage tanks. At 110 s, tank #5 in zone 5 starts a fire.

According to the types and storage of hazardous materials, the accident physical effect fields can be calculated by applying the accident analysis models of fire, explosion, and toxic release. Combined with the data of occurrence frequency, the dynamic individual risk distribution of the multi-hazard domino accident process can be calculated and serves as the risk field-perception basis for the evacuation path planning method, as shown in Fig. 4. The models and data used in this case study are presented in Appendix (Tables 1-4).

3.3. Comprehensive risk analysis

In addition to applying dynamic risk analysis as the risk field-perception basis for the evacuation path planning method, the case study also performs 10,000 single simulations of the multi-hazard accidents to conduct the MCS. The single simulation starts from a random primary accident at a random installation, and dynamically deduces the uncertain process of accident development. Calculation of the comprehensive risk distribution is based on the statistical averaging of the data from the MCS result. Fig. 5 shows the comprehensive individual risk distribution over time. Comprehensive risk analysis serves as the risk pre-perception basis for the evacuation path planning method.

3.4. Optimal path planning and CIR calculation

The starting point of the evacuation is located at the star position in Fig. 3. The case study sets two different types of evacuation targets for evacuation path planning. The first is to leave the area from an arbitrary boundary location. The second is to leave the area from three designated

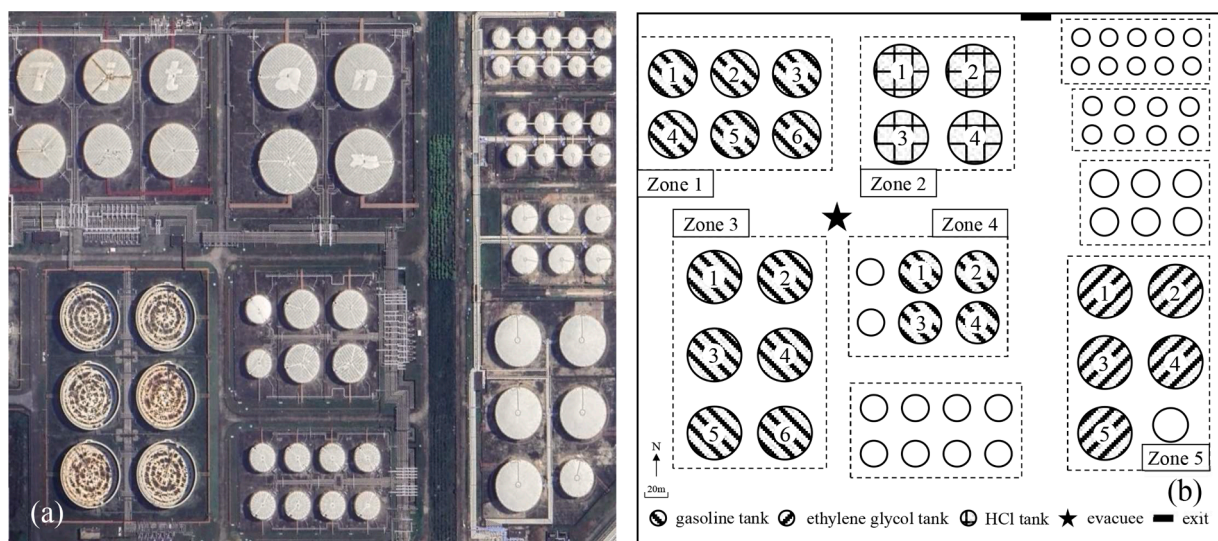


Fig. 3. (a) The overlook and (b) the plan of the chemical park.

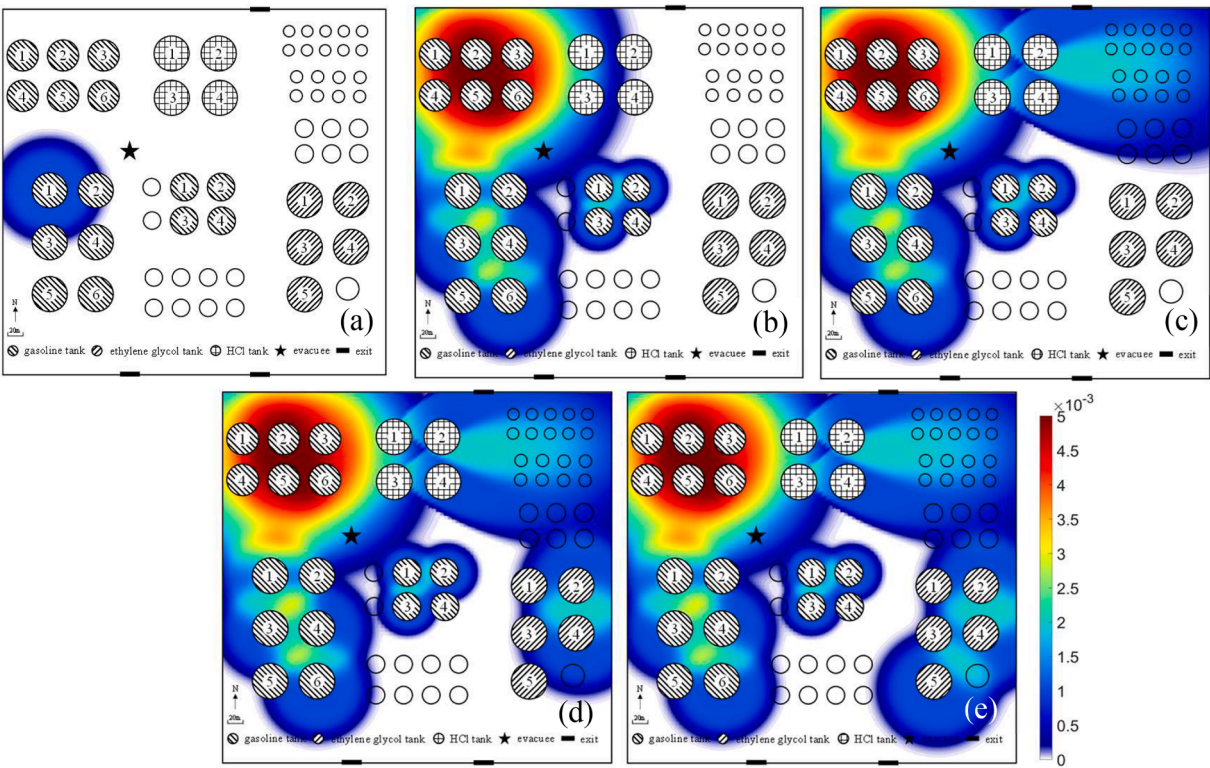


Fig. 4. Dynamic risk analysis results of the domino accident at (a) 1 second, (b) 10 s, (c) 38 s, (d) 92 s, and (e) 110 s.

Table 1
Installations and hazardous materials information.

	Hazardous material	Heat of combustion (H _c)	Burning rate (dm/dt, 10 ⁻³ kg/m ² ·s)	Capacity (kJ/kg·°C)	Density (g/mL)	Storage (kg)	Accident types
Zone 1 #1~#6	gasoline	44,000 kJ/kg	81	2.0	0.74	87,700	fire, explosion
Zone 2 #1~#4	hydrogen chloride (liquid)	/	/	1.18	1.03	15,000	toxic release
Zone 3 #1~#6	gasoline	44,000 kJ/kg	81	2.0	0.74	20,610	fire, explosion
Zone 4 #1~#4	gasoline	44,000 kJ/kg	81	2.0	0.74	5320	fire, explosion
Zone 5 #1~#5	ethylene glycol	1180 kJ/mol	0.034	2.4	1.11	57,700	fire, explosion

Table 2
Other information used in the case study.

Atmospheric stability	Wind speed (u)	Leakage speed (Q)	Wind direction	Toxic parameters	Air temperature	Humidity	MCS times
C	3 m/s	100 kg/s	west	A = -21.76; B = 2.65; n = 1	20 °C	20 %	10,000

exits (as shown in Fig. 3). The optimal evacuation path for each evacuation target is obtained by the improved D* algorithm. This section shows the results of optimal path planning corresponding to the D* algorithm based on two types of heuristic functions with different risk preferences ($\alpha = 1, \beta = 0$ and $\alpha = 0.5, \beta = 0.5$).

First, the case study tests the evacuation path planning algorithm whose heuristic function is independently determined by the dynamic risk analysis ($\alpha = 1, \beta = 0$), which means the risk field-perception completely determines the path planning, regardless of the pre-perception of risk. Fig. 6 shows the optimal evacuation paths planned according to this algorithm with the two different evacuation targets (red lines). In comparison, Fig. 6 also shows the evacuation paths planned by evacuees based on their subconscious decisions in the same accident scenario (green lines). For the first evacuation target, evacuees

without designated exits will subconsciously avoid high-risk areas during the evacuation process. The algorithm applies cellular automaton to simulate the risk-aversion preference of evacuees. For the second target, the subconscious decision of evacuees is to get to the nearest designated exit as quickly as possible. The algorithm uses Euclidean distance as the estimated cost to simulate the preference of evacuees.

Fig. 7 shows the CIR-time curves of the four evacuation paths in Fig. 6. CIR curves in Fig. 7(a) show that compared to the evacuees' subconscious path planning, the application of risk field-perception indicator as the equivalent distance in the heuristic function can reduce the CIR of the evacuation process from 0.097 to 0.0873, about 10 % lower. The contrast in Fig. 7(b) is more obvious. When evacuees regard the exits as their evacuation destination, the evacuation path planning method with the application of the equivalent distance improvement

Table 3
Models used in the dynamic risk analysis and comprehensive risk analysis.

	Accident types	Model types	Equations
consequence analysis models	fire	point source model	$I = \frac{H_c t_c \eta}{4\pi r^2} \frac{dm}{dt}$ (t_c -transmissivity=1, η -radiation fraction=0.13–0.35, r -distance (m))
	explosion	TNT-equivalent model	$P = 1.02 \frac{m_{TNT}^{1/3}}{r} + 3.99 \frac{m_{TNT}^{2/3}}{r^2} + 12.6 \frac{m_{TNT}}{r^3}$ (m_{TNT} -TNT-equivalent mass (kg), r -distance (m))
	toxic release	Gaussian plume model	$C = \frac{Q}{\pi \sigma_y \sigma_z u} e^{-\frac{y^2}{2\sigma_y^2}}$ $\sigma_y = 0.11x(1 + 0.0001x)^{-0.5}$ $\sigma_z = 0.08x(1 + 0.0002x)^{0.5}$ (x -axial distance (m), y -normal distance (m))
escalation probability models	thermal radiation	probit value model	$Y = 12.54 - 1.847(-1.128 \ln I - 2.667 \times 10^{-5} V + 9.877)$ (V -installation volume (m^3))
damage analysis models	overpressure	probit value model	$Y = -18.96 + 2.44 \ln P_r$
	thermal radiation	probit value model	$P_r = -14.9 + 2.56 \ln(6 \times 10^{-3} I^{1.33} t)$ (t -duration of exposure (s))
	overpressure toxic		$P_r = 5.13 + 1.37 \ln(P)$ $P_r = A + B \ln(C^a t)$

Table 4
Data of accident frequency used in the case study.

Accident types	Frequency as primary accident (y^{-1})
fire	5.6×10^{-4}
explosion	3.4×10^{-4}
toxic release	1×10^{-3}

significantly reduces the CIR of the evacuation process from 0.0947 to 0.0204, about 78.5 % lower. Although the evacuation takes more time, the risk to evacuees is lower.

The comparison between Fig. 7(a) and (b) shows the importance of emergency exit settings. Compared to the second evacuation target, the first target provides the evacuees with more alternative destinations, but it also makes the evacuees more confused in the decision-making process during the evacuation. This makes the CIR of the optimal evacuation path without designated exits four times greater than that with designated exits. The comparison shows that planning emergency exits and

evacuation paths in advance is important to improve evacuation efficiency.

The above two types of comparisons illustrate that avoiding blind evacuation can effectively improve evacuation efficiency in chemical accidents. In the 1984 toxic gas release accident in Bhopal, India, which killed 25,000 people, none of the factory workers closest to the accident center died. They were told to evacuate in the opposite direction of the toxic diffusion, thus quickly escaping from the high-risk area [36]. Based on reasonable exit settings and risk field-perception of the real-time development of the accident, the proposed evacuation path planning method is expected to significantly reduce the accident risk suffered by evacuees.

Further, the case study takes the comprehensive risk analysis results into consideration in the evacuation path planning method. The heuristic function of the path planning algorithm is adjusted to an equal proportion of field-perception and pre-perception of the accident risk ($\alpha, \beta = 0.5$), which means the optimal evacuation path is determined by both real-time accident scenario and risk prediction. Fig. 8(a) shows the two optimal evacuation paths with the target of leaving the area from any boundary location (green—risk field-perception only, red—risk field-perception and pre-perception). Fig. 8(b) shows the optimal evacuation paths ending at the designated exits.

This step further improves the evacuation route planning method. This improvement enables evacuation planners to have the ability to "predict" possible future accidents, and comprehensively determine the optimal evacuation path based on the real-time accident scenario and risk prediction. As shown in Fig. 8, the red lines have a more obvious risk aversion tendency than the green lines. Fig. 9 shows the CIR-time curves of the four evacuation paths in Fig. 8. The comparison of CIR curves in Fig. 9(a) shows the positive effect of risk pre-perception on improving evacuation efficiency. The CIR of the evacuation path drops from 0.0873 to 0.025, about 71.4 % lower. Evacuation time also decreases by about 24.7 %. Fig. 9(b) shows the optimization effect of the comprehensive risk perception on the evacuation paths with designated exits. The CIR of the evacuation path drops from 0.0204 to 0.0155, about 24 % lower. Evacuation time decreases by about 9.8 %.

The explosion accident that occurred in Yingkou, China in 2023 can be a confirmation of the case study. The nine casualties caused by the accident all occurred during the evacuation process. The leakage of high-temperature materials led to a secondary vapor explosion, whose location was close to the emergency evacuation path. One of the major causes of the accident was that the evacuation planner did not have a pre-perception of potential explosion accidents, resulting in unreasonable planning and design of the evacuation path [3]. The casualties may not be so serious if the evacuees could be pre-informed of the comprehensive risk of the area and avoid the high-risk areas in advance.

The case study section proves that the proposed evacuation path planning method for multi-hazard accidents can significantly optimize the evacuation efficiency. Compared to the subconscious decision made

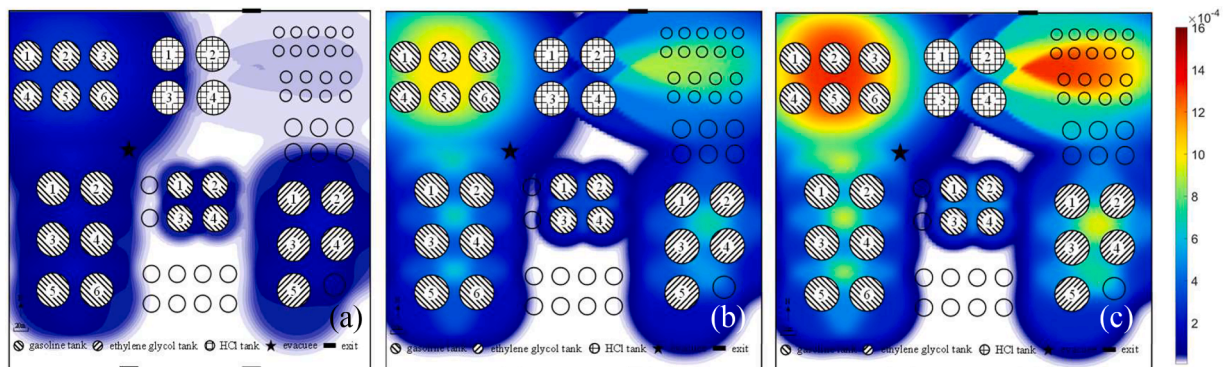


Fig. 5. Comprehensive risk analysis results of the accident at (a) 38 s, (b) 110 s, and (c) 200 s.

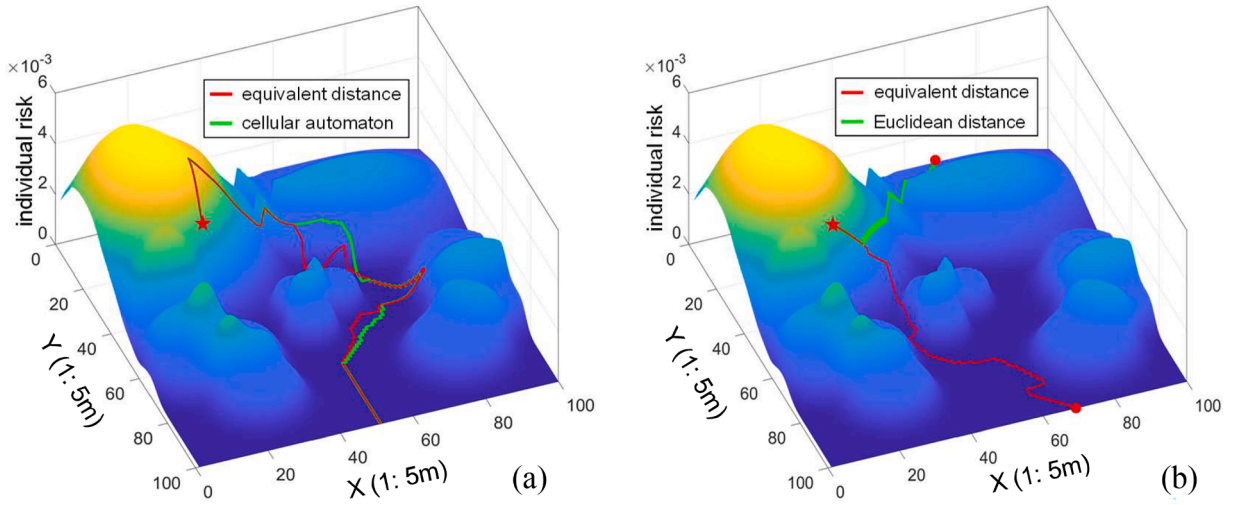


Fig. 6. Comparison of the optimal evacuation paths planned by the risk field-perception algorithm and the paths subconsciously decided by evacuees. (a) without designated exits, (b) with the three designated exits. (The 3-D background is the individual risk distribution at 110 s).

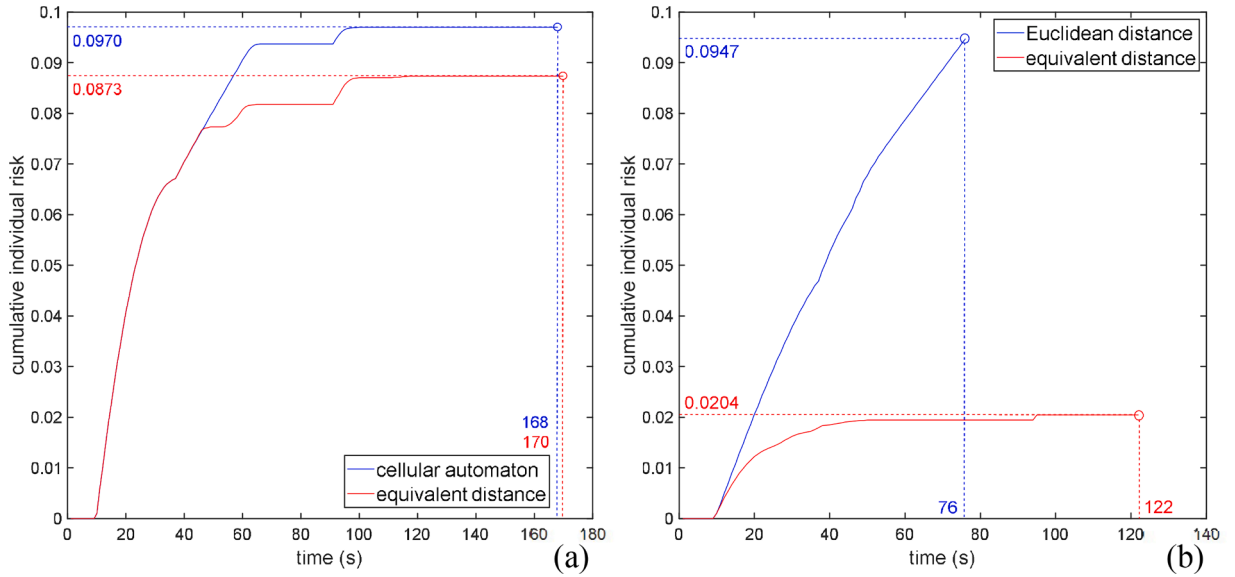


Fig. 7. The CIR-time curves of the evacuation paths planned by the risk field-perception algorithm and the paths subconsciously decided by evacuees. (a) without designated exits, (b) with the three designated exits.

by evacuees, this method applies risk field-perception and pre-perception to reduce the accident risk suffered by evacuees during the evacuation process by 74.2 % and 83.6 % respectively. The case study also proves the importance of setting reasonable evacuation exits to improve evacuation efficiency, and suggests that free but blind evacuation planning may actually increase the evacuation risk.

4. Discussion

The same to path planning in the transportation field, evacuation path planning in chemical accidents also faces the challenges of different actual situations and rapidly changing accident development. The challenges require the evacuation path planning method to have sufficient flexibility to consider the preferences of evacuation planners and different requirements of the application. This section aims to introduce the flexibility of the proposed evacuation path planning method from three aspects: environmental influence, evacuation preference, and emergency response, based on the case study.

4.1. Consideration of environmental factors

By changing the direction and speed of the diffusion of toxic gas, wind can significantly influence the consequences and risk of chemical accidents. The evacuation path planning method should be able to consider the influence of the wind and other environmental factors on the accident risk, including real-time and comprehensive risks, thereby determining the optimal evacuation path suitable for different accident scenarios. The wind direction in the case study is set to west, as shown in Figs. 4 and 5. This discussion changes the wind direction to northwest, north, and northeast, respectively, and re-calculates the risk analysis results accordingly. The optimal paths determined by the evacuation path planning method are shown in Fig. 10(a), and the CIR corresponding to each path is shown in Fig. 10(b).

Fig. 10(a) shows that as the wind direction changes, the evacuation paths with the designated exits change between the southeast and southwest exits as the destinations. Free evacuation paths without designated exits change irregularly. Fig. 10(b) shows that the CIRs of free evacuation are generally greater than the paths with designated

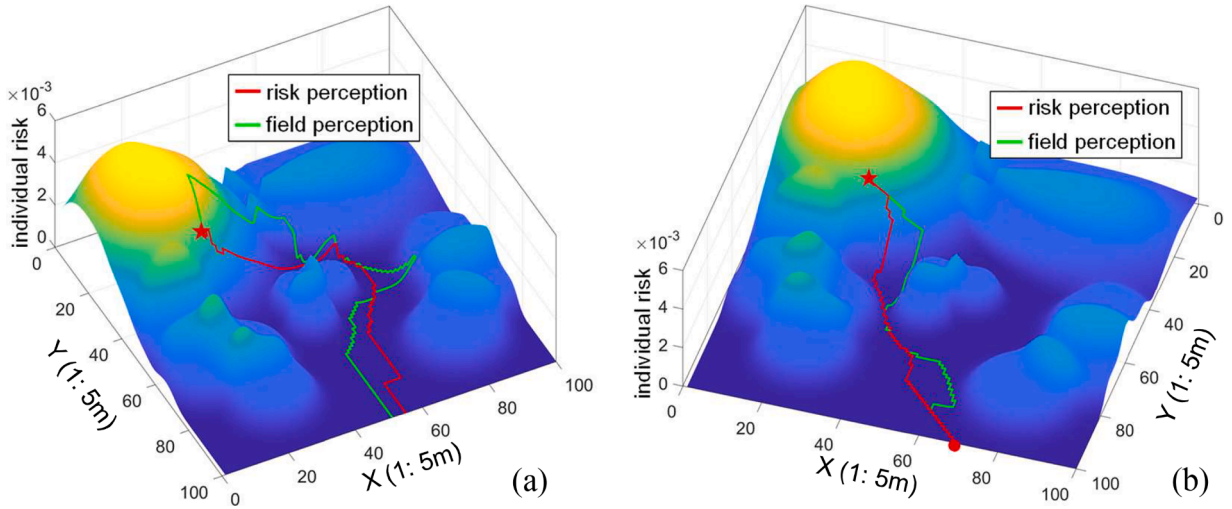


Fig. 8. Comparison of the optimal evacuation paths planned by the field-perception and pre-perception algorithm and the field-perception-only algorithm. (a) without designated exits, (b) with the three designated exits.

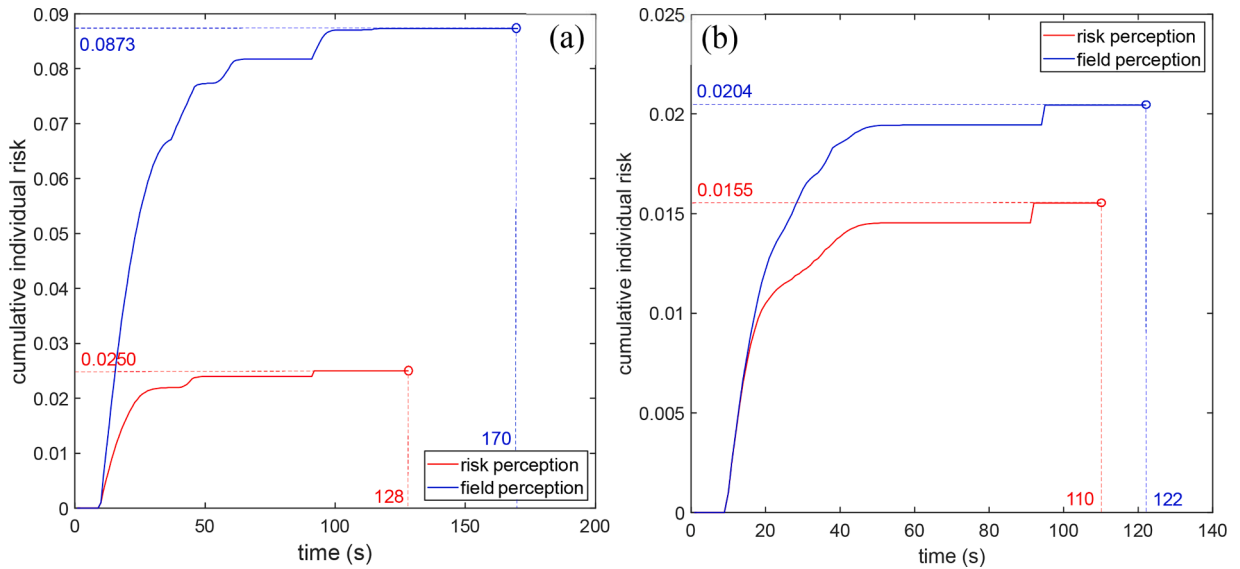


Fig. 9. The CIR-time curves of the evacuation paths planned by the field-perception and pre-perception algorithm and the field-perception-only algorithm. (a) without designated exits, (b) with the three designated exits.

exits, except for the evacuation path in the north wind. Under the influence of the north wind, the diffusion of toxic gas causes all exits to be in high-risk areas. The mandatory requirement to evacuate from the exits conversely increases the evacuation risk. This indicates that the setting of evacuation exits should be even-distributed and risk-based to be reasonable, in order to effectively improve the evacuation efficiency in accidents.

This discussion indicates that environmental factors such as wind direction can significantly influence the results of evacuation path planning in chemical accidents. The CIRs of evacuation paths influenced by wind direction can be increased by up to 3.4 times. Calculating the risk distribution of the dynamic development of the accident according to the real-time wind direction, and conducting the comprehensive risk analysis according to the prevailing wind direction in different periods of time in the area, are both essential support for effective evacuation path planning.

The well-known toxic release accident that happened in Seveso, Italy, in 1976, can be a confirmation of the conclusion of this discussion. The toxic gas spread with the 5 m/s southeast wind and affected

humans, animals, and the environment in an area of 1810 arcs downwind [37]. In addition to evacuation planning, safety management such as land-use planning and emergency response also need to consider environmental factors such as wind.

4.2. Evacuation preference

The applicability of an evacuation path planning method depends on whether it can be applied to various accident scenarios. Different chemical accident scenarios correspond to different environmental factors such as topography, landform, and meteorology, as well as different human factors such as evacuation planners' personalities, mentality, strategies, etc. For example, in specific accident scenarios, evacuees will tend to evacuate upwind to avoid exposure to the diffusion of toxic gas, and to higher ground to avoid the spread of leaked hazardous materials. In accidents with expectedly slow escalation and development, evacuees prefer to leave the industrial area in the shortest time.

The proposed evacuation path planning method can consider different evacuation preferences in various accident scenarios with the

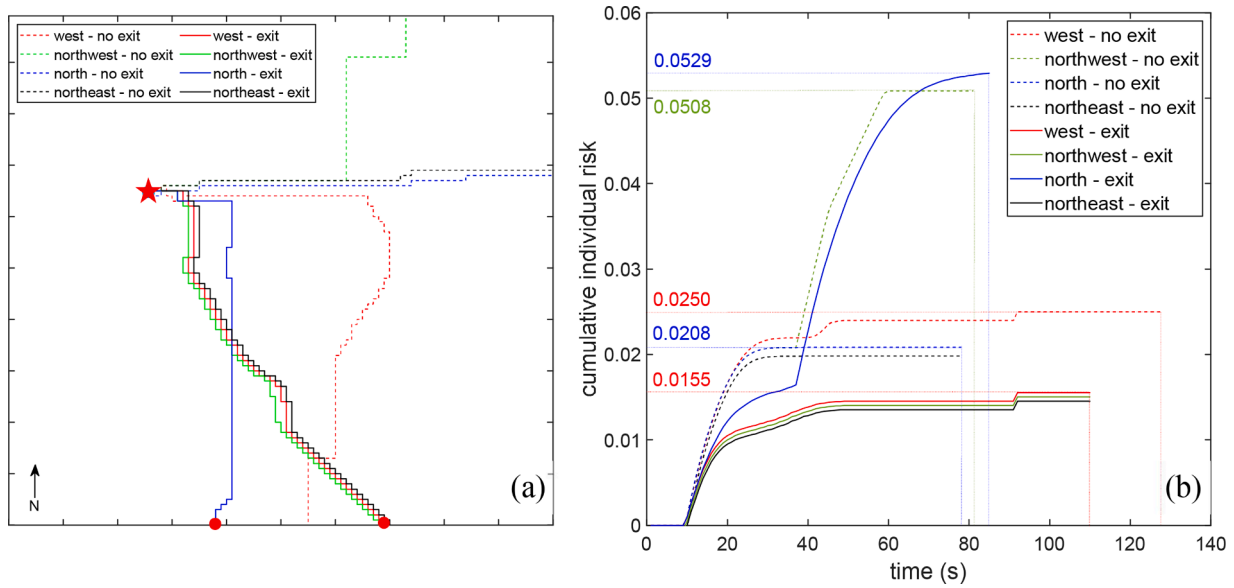


Fig. 10. (a) The optimal evacuation paths in different wind directions and (b) the corresponding CIRs.

improved D* algorithm by adjusting the weights of the heuristic function and adding new preference terms. The discussion section shows the evacuation path planning results of the method considering timesaving preference and direction preference respectively.

First, a time-consumed evaluation term is added to the heuristic function of the path planning algorithm, which has the same weight as the risk indicator. It should be noted that due to the non-additivity between the two indicators of risk analysis and time consumed evaluation, the normalization of each evaluation term should be the basis of the addition of new preference terms. Fig. 11(a) shows the evacuation path planning results considering the timesaving preference, and the comparison to the original results with the risk-only evaluation indicators. Fig. 11(b) shows the CIRs corresponding to different evacuation paths.

Fig. 11 shows that after the introduction of the timesaving preference terms, evacuees make different path choices. The CIRs corresponding to the optimal paths of exit-fixed and free evacuation increase from 0.0155 to 0.0383 and from 0.025 to 0.0312, increasing by 147 and 24.8 %, respectively.

However, the addition of the preference term does not have the expected effect of reducing the evacuation time consumed. As shown in Fig. 11(b), the time consumed of free evacuation decreases from 128 s to 116 s, merely 9.4 %. The time consumed of exit-fixed evacuation even increases from 110 s to 118 s, about 7.3 % longer. As shown in Fig. 11(a), the evacuees with timesaving preference choose to move closer to the evacuation destination after the accident occurs. However, secondary accidents during the evacuation process force them to re-plan their evacuation path. The case indicates that in accident scenarios that develop rapidly and unpredictably, blindly preferring the speed of evacuation may conversely increase the time consumed and evacuation risk.

Furthermore, two direction preference terms are also added to the heuristic function of the path planning algorithm, to upwind (west) and upland (east) directions respectively. The direction preference terms have the same weights as the risk indicator. Fig. 12(a) shows the evacuation paths planned with the two direction preference terms. Fig. 12(b)

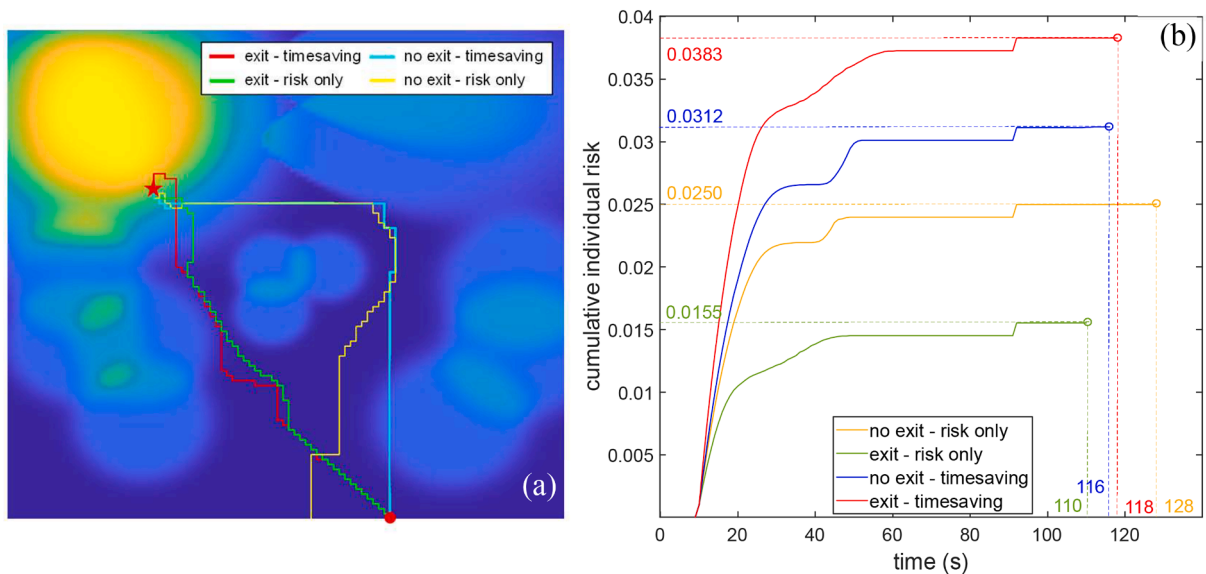


Fig. 11. (a) The comparison between optimal evacuation paths with and without timesaving preference and (b) the CIRs of the evacuation paths.

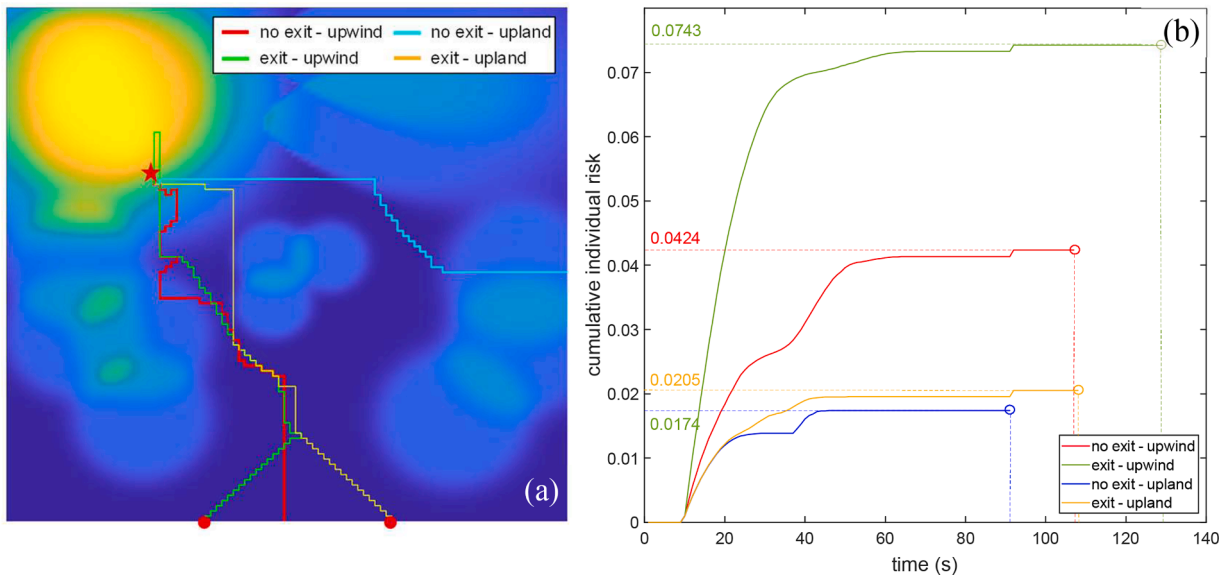


Fig. 12. (a) The comparison between optimal evacuation paths with the two direction preferences and (b) the CIRs of the evacuation paths.

shows the corresponding CIRs of the paths.

Compared to Fig. 8, the consideration of the direction preferences significantly influences the optimal paths planned by the method. The evacuation paths shown in Fig. 12(a) generally increase the CIRs of the exit-fixed and free evacuation while attempting to satisfy the direction preferences. The upwind preference increases the CIRs for exit-fixed and free evacuation from 0.0155 to 0.025 to 0.0734 and 0.0424, 373.5 and 69.6 % higher respectively. The upwind preference increases the CIR for exit-fixed evacuation from 0.0155 to 0.0205, 32.3 % higher.

There are two anomalies in Fig. 12 that deserve further discussion. The first is that under the influence of direction preferences, the evacuation risks of exit-fixed evacuation are instead higher than free evacuation. This phenomenon confirms the importance of properly setting up emergency exits. Exit settings should comprehensively consider the actual on-site situation and evacuees' general preferences, otherwise may reduce the evacuation efficiency instead.

The second is that the CIR of free evacuation with upland preference (0.0174) is lower than that with risk-aversion preference only (0.025). This is because the high-risk areas in the comprehensive risk analysis, which is the component of the heuristic function, do not mean that the accident will occur at the locations immediately. When the evacuees in Fig. 12(a) follow the blue line to cross the high-risk area on the east side (before 92 s), the accidents in this area have not occurred yet (occur after 92 s). This phenomenon indicates that risk pre-perception can make evacuation path planning relatively more efficient, but excessive consideration of the comprehensive risk may make evacuees hesitate and miss better escape options. Reasonably adjusting the weight of each preference term in the heuristic function is an important decision that evacuation planners need to make according to the actual situation.

4.3. Strategy for emergency response

In addition to emergency evacuation, emergency response is also an important part of emergency management of chemical accidents. After an accident occurs, emergency responders (fire brigade, medical staff, etc.) need to determine the emergency response strategy based on the on-site accident scenario and the prediction of the dynamic development of the accident. The flexibility of the proposed evacuation path planning method enables it to be applied to the optimization of the emergency response strategy. This section aims to introduce the planning of emergency response approach paths by applying the proposed method based on risk preference and timesaving preference.

The major modification of the path planning algorithm for emergency response is the inversion of the real-time and comprehensive risk distributions according to the timeline. Then the destination of the emergency response can be regarded as the starting point of the evacuation planning algorithm. The improved D* algorithm is applied to calculate the locations that the path can reach at different times according to different risk distributions, until the location coincides with the starting point of the emergency response.

Fig. 13(a) shows the optimal approach paths for emergency response planned by the modified method with and without designated entrance. Fig. 13(b) shows the corresponding CIRs of the two paths. The applied algorithm aims at minimizing the risk of emergency response, whose heuristic function is composed of risk field-perception and pre-perception in equal proportions. As shown in Fig. 13(b), the CIR of the emergency response process is smaller than that of the evacuation process. The reason is that emergency responders at the edge of the area suffer less accident risk at the beginning of the accident compared to evacuees.

However, the emergency response strategy is more aimed at minimizing the time consumed instead of the individual risk. Fig. 14(a) shows the optimal approach paths planned by adding the timesaving preference term into the path planning algorithm and adjusting its weight in the heuristic function to 50 and 80 %, respectively. Fig. 14(b) shows the corresponding CIRs of the paths. The CIR curves show that the timesaving terms effectively shorten the approach time consumed from 89 s to 33 s (50 %) and 27 s (80 %) (free approach, up to 69.7 % shorter), and from 111 s to 79 s (50 %) and 75 s (80 %) (entrance-fixed approach, up to 32.4 % shorter). In comparison, the effect of time-saving preference is more obvious in the free emergency approach. For emergency responders such as firefighters who require the shortest time consumed, the approach path without designated entrances can be more efficient.

Fig. 14(b) also shows that the timesaving preference significantly increases the CIRs of emergency response, from 0.016 to 0.0681 (free approach, 325.6 % higher), and from 0.0118 to 0.0273 (entrance-fixed approach, 131.4 % higher). However, professionally equipped emergency responders have higher risk tolerance than evacuees. Determining the risk tolerance limit of emergency responders and minimizing the time consumed of the approach path within the limit, is an optimization problem that emergency response planners can solve by applying the proposed method.

This discussion indicates that efficient emergency response also requires risk field-perception and pre-perception of the area as the

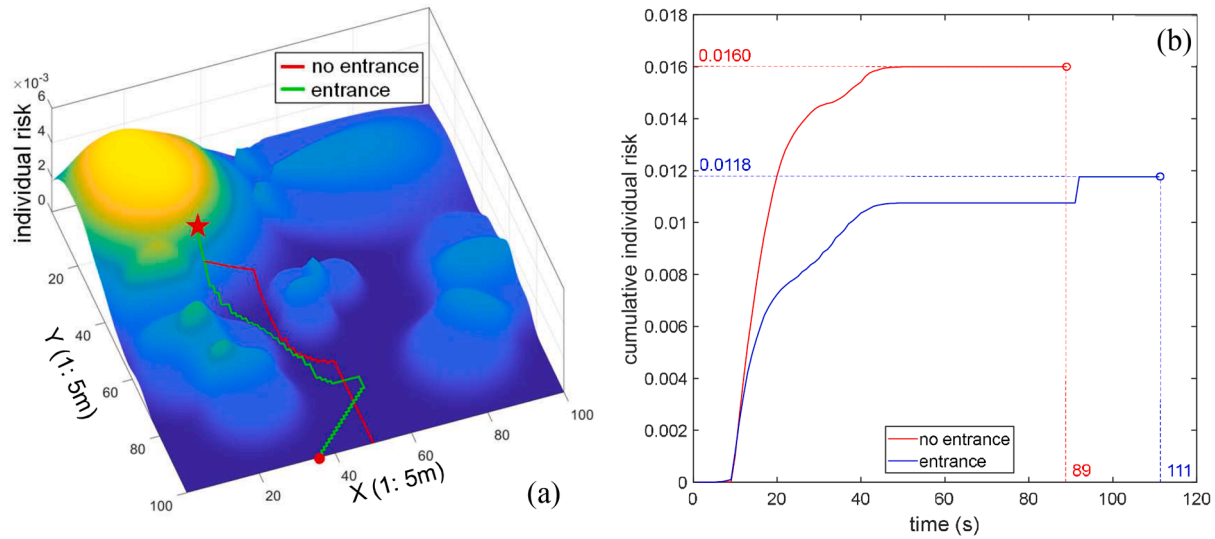


Fig. 13. (a) The optimal approach paths for emergency response based on risk perception and (b) the CIRs of the approach paths.

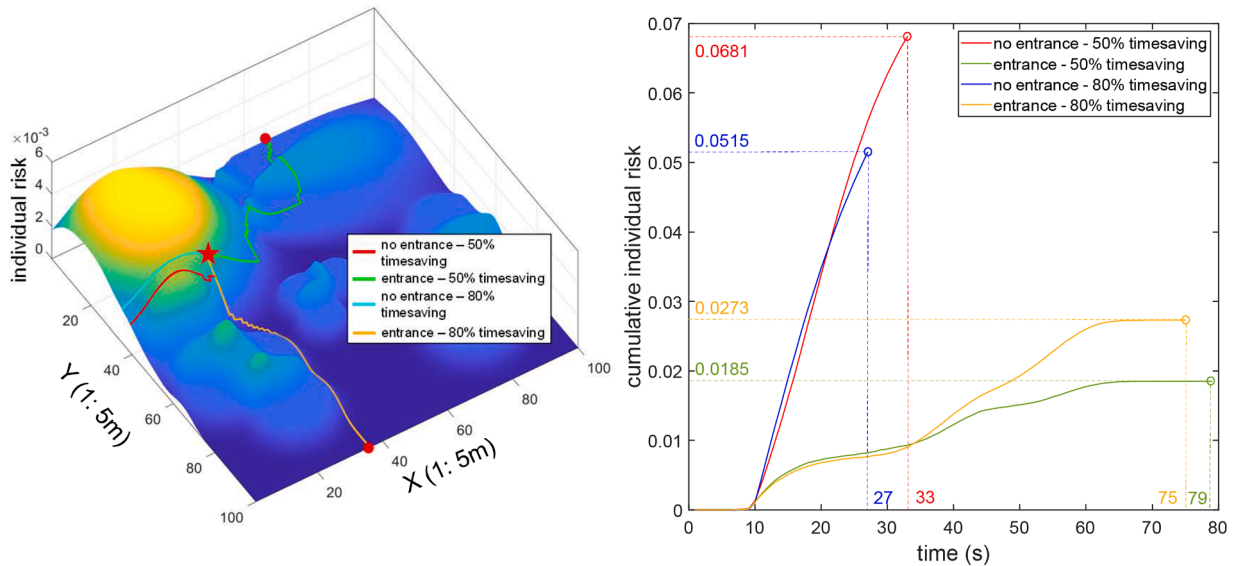


Fig. 14. (a) The optimal approach paths for emergency response considering timesaving preferences and (b) the CIRs of the approach paths.

support. The multi-hazard accident that occurred in Tianjin, China, in 2015 is a typical negative case. After the occurrence of the fire accident, the fire brigade that first entered the accident scene was not informed that the explosive ammonium nitrate was stored in the accident area. The subsequent explosion killed all 25 members of the team [38].

The consideration of different evacuation preferences and the application in emergency response strategy optimization of the path planning method both benefit from the improved D* algorithm. With the flexible heuristic function, the improved D* algorithm allows the optimization criteria of path planning to be adjusted according to actual needs. This is the major advantage and innovation of the proposed evacuation path planning method.

5. Conclusion

With the agglomerative and large-scale development of chemical industries, the frequency of multi-hazard chemical accidents including fires, explosions, and toxic releases has been continually increasing.

Safety management strategies in chemical industries should be improved in response to the uncertain, unpredictable, and nonlinear characteristics of multi-hazard accidents. Evacuation planning is an important research topic in accident emergency management. Improving the evacuation path planning algorithm to enhance its applicability in dynamic and multi-hazard chemical accidents has important research value and practical significance.

This paper presents an evacuation path planning method for multi-hazard accidents in chemical industries based on advance and on-site risk identification and assessment. This method is based on the improved D* path planning algorithm, which simulates the evacuee's prediction and real-time perception of accident risks, and applies the concept of equivalent distance to improve the heuristic function. The heuristic function in this method consists of the cumulative individual risks calculated by the dynamic risk analysis and comprehensive risk analysis, which are respectively based on the dynamic simulation and analysis of the accident development, and the data synthesis and statistical analysis applying the Monte Carlo simulation. This method aims

to simultaneously consider the comprehensive factors including time consumed, real-time risk, and risk prediction in the optimal evacuation path planning criteria.

A chemical park in Guangzhou, China is adopted as the case study to verify the effectiveness of the proposed evacuation path planning method. The case study quantitatively demonstrates the effect of this method on improving evacuation efficiency in accidents. Based on the dynamic and comprehensive risk analysis of the multi-hazard accident, the proposed method determines the optimal evacuation paths in free and exit-fixed evacuations. The proposed method can effectively reduce the evacuation risk during the free and exit-fixed evacuation process, compared to the evacuation decisions made by evacuees based on subconsciousness, by 74.2 and 83.6 % respectively.

This paper further discusses the flexibility of the proposed evacuation path planning method. The first discusses that the method can consider the influence of wind and other environmental factors, and demonstrates the influence of the wind direction on the evacuation path planning results; The second, by introducing the timesaving and direction preference terms, the discussion section demonstrates the flexibility of the proposed method in considering different evacuation preferences; The third discusses the applicability of the proposed method in the optimization of emergency response strategies; The discussion section further proves that reasonable emergency exit settings can prevent evacuees from being confused and blind during the evacuation, thereby improving evacuation efficiency.

The evacuation path planning method proposed in this paper is expected to contribute to the construction of the emergency management framework for multi-hazard chemical accidents. The method can provide effective tools for the enhancement of the safety of chemical industries, and provide a reference for policymakers to improve the loss prevention policies for multi-hazard accidents in chemical industries. However, **the disadvantages of this method are that it cannot consider the crowding and congestion problems in the evacuation of dense crowds, and it lacks the consideration of the damage caused by the accident to evacuees. In-depth research and solving the above shortcomings will be the future goal of the optimization of this method.**

CRediT authorship contribution statement

Zhichao He: Writing – original draft, Software, Methodology, Investigation, Conceptualization. **Kaixin Shen:** Writing – review & editing, Methodology, Conceptualization. **Meng Lan:** Writing – review & editing, Conceptualization. **Wenguo Weng:** Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare no conflict of interests.

Acknowledgments

This study was supported by the National Natural Science Foundation of China (Grant No. 72034004), the National Key Research and Development Program of China (Grant Nos. 2023YFC3807602, 2023YFC3008702, 2023YFC3008703), and Postdoctoral Researcher Funding Program of China (Grant No. GZB20230359). The authors are deeply grateful to these supports.

Appendix: Data and models applied in the case study

This appendix aims to provide data used in the dynamic risk analysis and comprehensive risk analysis conducted in the case study section, including hazardous materials, installations, environmental information, etc., as well as the applied consequence analysis, escalation probability, damage analysis models, and accident frequency data required for the risk analysis.

It should be noted that in order to reduce the complexity of the accident scene, the case study adopted reasonable assumptions and simplifications for the data and models. The case study aims to verify the effectiveness of the proposed evacuation path planning method in an actual chemical park. The constructed accident scenarios and applied data may be different from the actual on-site situation.

References

- [1] He Z, Chen C, Weng W. Multi-hazard risk assessment in process industries: state-of-the-art. *J Loss Prev Process Ind* 2021;76:104672. <https://doi.org/10.1016/j.jlp.2021.104672>.
- [2] MEM (Ministry of Emergency Management of the People's Republic of China). Investigation report on “3-21” explosion accident of Tianjiayi chemical co., Ltd. Xiangshui, Jiangsu; 2019 (accessed November 2019). <https://www.mem.gov.cn/gk/sgcc/tbzdsgdcbg/2019tbzdsgcc/201911/P02019111556511829069.pdf>.
- [3] MEM (Ministry of Emergency Management of the People's Republic of China). An alkylation unit exploded at a company in Panjin. Liaoning; 2023 (accessed January 2023). https://www.mem.gov.cn/xw/bndt/202301/t20230116_440661.shtml.
- [4] Wang J, Yu X, Zong R, et al. Evacuation route optimization under real-time toxic gas dispersion through CFD simulation and Dijkstra algorithm. *J Loss Prev Process Ind* 2022;76:104733. <https://doi.org/10.1016/j.jlp.2022.104733>.
- [5] Gai WM, Deng YF, Jiang ZA, et al. Multi-objective evacuation routing optimization for toxic cloud releases. *Reliab Eng Syst Saf* 2017;159:58–68. <https://doi.org/10.1016/j.res.2016.10.021>.
- [6] Chen G, Liang X, Zhou L, et al. A dynamic risk-based routing approach for multi-source and multi-sink evacuation problem in chemical industrial parks. *J Loss Prev Process Ind* 2022;77:104760. <https://doi.org/10.1016/j.jlp.2022.104760>.
- [7] Li JJ, Zhu HY. A risk-based model of evacuation route optimization under fire. *Procedia Eng* 2018;211:365–71. <https://doi.org/10.1016/j.proeng.2017.12.024>.
- [8] Gai W, Deng Y. *Emergency guidance methods and strategies for major chemical accidents*. Singapore: Springer Nature Singapore; 2022.
- [9] Dou X, Mebarki A, Cheng Y, et al. Review on the emergency evacuation in chemicals-concentrated areas. *J Loss Prev Process Ind* 2019;60:35–45. <https://doi.org/10.1016/j.jlp.2019.03.008>.
- [10] Seo SK, Yoon YG, Lee JS, et al. Deep neural network-based optimization framework for safety evacuation route during toxic gas leak incidents. *Reliab Eng Syst Saf* 2022;218:108102. <https://doi.org/10.1016/j.res.2021.108102>.
- [11] Li S, Ma H, Zhang Y, et al. Emergency evacuation risk assessment method for educational buildings based on improved extreme learning machine. *Reliab Eng Syst Saf* 2023;238:109454. <https://doi.org/10.1016/j.res.2023.109454>.
- [12] Wang X, Xia G, Zhao J, et al. A novel method for the risk assessment of human evacuation from cruise ships in maritime transportation. *Reliab Eng Syst Saf* 2023;230:108887. <https://doi.org/10.1016/j.res.2022.108887>.
- [13] Yuan S, Cai J, Reniers G, et al. Safety barrier performance assessment by integrating computational fluid dynamics and evacuation modeling for toxic gas leakage scenarios. *Reliab Eng Syst Saf* 2022;226:108719. <https://doi.org/10.1016/j.res.2022.108719>.
- [14] Yoo B, Choi SD. Emergency evacuation plan for hazardous chemicals leakage accidents using GIS-based risk analysis techniques in South Korea. *Int J Environ Res Public Health* 2019;16(11):1948. <https://doi.org/10.3390/ijerph16111948>.
- [15] Gai WM, Du Y, Deng YF. Evacuation risk assessment of regional evacuation for major accidents and its application in emergency planning: a case study. *Saf Sci* 2018;106:203–18. <https://doi.org/10.1016/j.ssci.2018.03.021>.
- [16] Li C, Hou J, Cheng W, et al. A multiagent-based modeling approach for emergency evacuation plan optimization during toxic gas releases within chemical plants. *Process Saf Environ Protect* 2022;163:543–57. <https://doi.org/10.1016/j.psep.2022.05.026>.
- [17] Khakzad N. A goal programming approach to multi-objective optimization of firefighting strategies in the event of domino effects. *Reliab Eng Syst Saf* 2023;239:109523. <https://doi.org/10.1016/j.res.2023.109523>.
- [18] Khakzad N. A methodology based on Dijkstra's algorithm and mathematical programming for optimal evacuation in process plants in the event of major tank fires. *Reliab Eng Syst Saf* 2023;236:109291. <https://doi.org/10.1016/j.res.2023.109291>.
- [19] Choi M, Park M, Lee HS, et al. Understanding the role of dynamic risk perception during fire evacuations using agent-based modeling. *Construction Research Congress* 2016:1669–79. <https://doi.org/10.1061/9780784479827.167>.
- [20] Chen P, Chen G, Wang L, et al. Optimizing emergency rescue and evacuation planning with intelligent obstacle avoidance in a chemical industrial park. *J Loss Prev Process Ind* 2018;56:119–27. <https://doi.org/10.1016/j.jlp.2018.08.006>.
- [21] Onelcin P, Mutlu MM, Alver Y. Evacuation plan of an industrial zone: case study of a chemical accident in Aliaga, Turkey and the comparison of two different simulation softwares. *Saf Sci* 2013;60:123–30. <https://doi.org/10.1016/j.ssci.2013.07.006>.
- [22] Stentz A. Optimal and efficient path planning for partially-known environments. In: *Proceedings of the IEEE international conference on robotics and automation*; 1994. p. 3310–7. <https://doi.org/10.1109/ROBOT.1994.351061>.
- [23] Liu Q, Guo K, Wu X, et al. Simulation-based rescue plan modeling and performance assessment towards resilient metro systems under emergency. *Reliab Eng Syst Saf* 2024;241:109642. <https://doi.org/10.1016/j.res.2023.109642>.

- [24] Chen J, Zhong M, Qiu P, et al. Mapping the fire risk in buildings: a hybrid method of ASET-RSET concept and FED concept. *Reliab Eng Syst Saf* 2023;240:109571. <https://doi.org/10.1016/j.res.2023.109571>.
- [25] Reis MF. Solid waste incinerators: health impacts. *Encycl Environ Health* 2011: 162–217. <https://doi.org/10.1016/B978-0-444-52272-6.00489-X>.
- [26] Ferguson D, Stentz A. The delayed D* algorithm for efficient path replanning. In: *Proceedings of the IEEE international conference on robotics and automation*; 2005. p. 2045–50. <https://doi.org/10.1109/ROBOT.2005.1570414>.
- [27] Chen T, Zhang G, Hu X, et al. Unmanned aerial vehicle route planning method based on a star algorithm. In: *Proceedings of the 13th IEEE Conference on Industrial Electronics and Applications (ICIEA)*; 2018. p. 1510–4. <https://doi.org/10.1109/ICIEA.2018.8397948>.
- [28] Camillo A, Guillaume E, Rogaume T, et al. Risk analysis of fire and evacuation events in the European railway transport network. *Fire Saf J* 2013;60:25–36. <https://doi.org/10.1016/j.firesaf.2013.04.004>.
- [29] Fox MA, Spicer K, Chosewood LC, et al. Implications of applying cumulative risk assessment to the workplace. *Environ Int* 2018;115:230–8. <https://doi.org/10.1016/j.envint.2018.03.026>. *Environ. Int.*
- [30] Huang H, Wang A, Morello-Frosch R, et al. Cumulative risk and impact modeling on environmental chemical and social stressors. *Curr Environ Health Rep* 2018;5: 88–99. <https://doi.org/10.1007/s40572-018-0180-5>.
- [31] CCPS (Centre for Chemical Process Safety). *Guidelines for chemical process quantitative risk analysis, second edition*. New York: American Institute of Chemical Engineers; 2000.
- [32] TNO (Netherlands Organisation for Applied Scientific Research). *Methods for the calculation of physical effects, CPR 14E*. The Hague: Committee for the Prevention of Disasters; 1997.
- [33] He Z, Weng W. Synergic effects in the assessment of multi-hazard coupling disasters: fires, explosions, and toxicant leaks. *J Hazard Mater* 2020;388:121813. <https://doi.org/10.1016/j.jhazmat.2019.121813>.
- [34] He Z, Weng W. A dynamic and simulation-based method for quantitative risk assessment of the domino accident in chemical industry. *Process Saf Environ Protect* 2020;144:79–92. <https://doi.org/10.1016/j.psep.2020.07.014>.
- [35] Cozzani V, Gubinelli G, Antonioni G, et al. The assessment of risk caused by domino effect in quantitative area risk analysis. *J Hazard Mater* 2005;127(1–3): 14–30. <https://doi.org/10.1016/j.jhazmat.2005.07.003>.
- [36] Bowonder B. The Bhopal accident. *Technol Forecast Soc Change* 1987;32(2): 169–82. [https://doi.org/10.1016/0040-1625\(87\)90038-2](https://doi.org/10.1016/0040-1625(87)90038-2).
- [37] Homberger E, Reggiani G, Sambeth J, et al. The Seveso accident: its nature, extent and consequences. *Ann Occup Hyg* 1979;22(4):327–70. <https://doi.org/10.1093/annhyg/22.4.327>.
- [38] MEM (Ministry of Emergency Management of the People's Republic of China). Investigation report on “8-12” fire and explosion accident of Ruihai chemical warehouse. China: Tianjin; 2016 (accessed January 2021), <https://www.mem.gov.cn/gk/sgcc/tbzdsgdcbg/2016/201602/P020190415543917598002.pdf>.